Measurement and Uncertainty: Identifying Four Metals by Density

SYNOPSIS

You will determine the density of four metal rods twice — first with an imprecise set of tools, then with a precise set — and use each density to identify the metal. The physics here is simple; the skill being practiced is not the physics. This lab is about recording every measurement to the precision of the device that made it, reporting each value with the correct significant digits and its uncertainty, and judging a result by its percent error.

Due See the Module 1 Lab folder on Brightspace

Pairs with M1 N3 and M1 N4

Report Full lab report — use the Lab Template

Data Your own — four rods, two tool sets, three trials

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PERFORMANCE EXPECTATIONS By the end of this lab, you will be able to…

- **PE1** **Read a device with tick marks** — a ruler or a spring scale — down to the smallest tick, estimate one more digit between the ticks, and give the reading an uncertainty of half the smallest division, rounded to match the reading.
- **PE2** **Read a Vernier caliper and a digital scale** exactly as they resolve the reading, without guessing, and give each reading its uncertainty.
- **PE3** **Report every measured value with its uncertainty** — for example 3.30 ± 0.05 cm — and with the correct significant digits.
- **PE4** **Compute a volume, a density and a percent error** from averaged measurements, and show one sample calculation of each.
- **PE5** **Identify a metal from its density**, using the percent error and the rod's appearance, and apply the 10% rule.
- **PE6** **Compare two tool sets** using your own numbers: which was more precise, which gave the more confident identification, and whether the extra time was worth it.

📏 The one idea that matters: the device sets the uncertainty

Where the last digit of your reading comes from depends on the kind of device you are holding.

📐

Devices with tick marks — you estimate the last digit

Read down to the smallest tick mark, then estimate one more digit between the tick marks; that estimated digit is the last significant figure of your measurement. On a centimeter ruler with millimeter tick marks, an edge that falls between 12.4 cm and 12.5 cm might be recorded as 12.43 cm — the 3 is your estimate, and the uncertainty is half the smallest division, ±0.05 cm. The spring scale works exactly the same way: with tick marks every 5 g, a needle sitting between 45 g and 50 g might be recorded as 47 g. Half the 5 g interval is 2.5 g, which rounds to ±3 g so the uncertainty has the same decimal places as the reading: 47 ± 3 g. This is a general rule for anything with tick marks — rulers, spring scales, protractors, analog meters.

🔬

The Vernier caliper and the digital platform scale — no guessing

These devices resolve the last digit for you, so you simply read the value and write it down exactly as given; the uncertainty is the smallest interval the device resolves. The caliper reads to 1/50 mm = 0.002 cm, so a reading of 2.540 cm carries ±0.002 cm; the digital scale reads to 0.01 g, so a reading of 46.73 g carries ±0.01 g.

Compare the spring scale's ±3 g with the digital scale's ±0.01 g — that factor of 300 is exactly what “the precision of the device” means.

Report every measured value with its uncertainty, e.g. 3.30 ± 0.05 cm — and note the decimal places of the value and the uncertainty must match. We say “uncertainty” rather than “error”: nothing is wrong with these measurements; the device simply cannot tell us more.

Predict first. The end of a rod falls between the 12.4 cm and 12.5 cm marks on a ruler with millimeter ticks, a little less than halfway. How do you record the length?

(A) 12.4 cm

(B) 12.43 cm

(C) 12.430 cm

(D) 12 cm

What you need

- Four metal rods.
- A metric ruler with millimeter divisions.
- A hanging spring scale with a piece of thread.
- A Vernier caliper, and the handout on reading a Vernier scale (in **Files for this lab** below).
- A platform scale.
- The Excel data sheet for this lab (in **Files for this lab** below).
- The **Lab Template** and the **Writing Lab Reports** module on Brightspace. Read both before you start writing.

Part I — Ruler and hanging spring scale

1. For each of the four rods, measure the **length**, the **diameter**, and the **mass** three times each.
2. Hang the rod from the spring scale with the thread to read its mass.
3. **Do not repeat the same measurement three times in a row.** Cycle through length, diameter, mass, then repeat, so a mistake does not get copied three times.
4. Enter the raw readings into the Excel sheet (left-hand tables), which averages the trials for you.

Part II — Vernier caliper and platform scale

1. Review the handout on reading a Vernier scale first.
2. Then repeat all of Part I with the caliper and the platform scale, entering the readings into the right-hand tables.

Your caliper readings should carry more digits than your ruler readings — if they don't, you are not using the

Vernier scale.

Check yourself. Your ruler gives a rod's diameter as 1.27 cm. Your caliper reading of the same rod is also written as 1.27 cm. What does that tell you?

(A) The two tools agree, so both readings are good.

(B) The rod is very uniform.

(C) The caliper was read without its Vernier scale — it should give a third decimal place.

(D) The ruler reading must be wrong.

How to record each device

Table 1 — the four devices, from the examples above

Device	How you read the last digit	Example reading	Uncertainty
Ruler (mm ticks)	Estimate one digit between the ticks	12.43 cm	±0.05 cm (half a tick)
Hanging spring scale (5 g ticks)	Estimate one digit between the ticks	47 g	±3 g (half a tick, rounded to match the reading)
Vernier caliper	Read as given, from the Vernier scale	2.540 cm	±0.002 cm (one caliper step)
Digital platform scale	Read as given	46.73 g	±0.01 g (one display step)

Data analysis

From your averages, the sheet computes each rod's volume and density and suggests which metal your density is closest to. Density is mass per unit volume, and each rod is a cylinder:

$$\rho = m / V \qquad V = \pi r^2 L$$

r is the radius (half the diameter), L the length, m the mass

Because density is characteristic of a material, a measured density compared against a table of accepted values identifies the metal — and the agreement is quantified by the percent error:

$$\% \text{ error} = | \rho_{\text{measured}} - \rho_{\text{accepted}} | / \rho_{\text{accepted}} \times 100\%$$

1. Treat the sheet's suggestion as a **candidate, not an answer**.
2. Look up the accepted density of that metal in a reputable table, record where you found it (you will cite it), and type the accepted value into the **yellow cell**. The sheet then computes your percent error, below each table.
3. Decide which metal each rod is, using the percent error **and** the rod's appearance.

> **Check yourself. A rod's density comes out 8.71 g/cm³. The sheet's typical values list brass at 8.5 g/cm³ and copper at 8.96 g/cm³. What should you do?**
>
> **(A)** Call it brass and stop — it is the closer value.
>
> **(B)** Compute the percent error against both, look at the rod's color, and argue from both which metal it is.
>
> **(C)** Call it copper — copper is the more common metal.
>
> **(D)** Report it as unidentified.

What your report must contain

Write the report on the Lab Template, following the lab-writing guide (**How to Write a Lab Report** in the Writing Lab Reports module). Quote every directly measured value with the correct significant digits and its uncertainty, and include **one sample calculation each** for an average, a volume, a density, and a percent error in your Results section.

Questions for your discussion

Work these into your **Discussion** and **Additional Questions** sections, in sentences and paragraphs — never as a list. Apply the 10% rule when discussing your errors.

I. Which tool set gave the more precise measurements, and how do you know from the numbers?

II. Did the more precise tools give the smaller percent error and the more confident identification?

III. How did the measurement time compare between the two tool sets?

IV. Was the extra time worth it?

Reminders: sentences and paragraphs only — no bullet points or numbered lists anywhere in the report. Never write “human error.” Your references must include the source of your density table.

✔ Before you submit

Every group here is a row on the Lab 01 grading rubric.

ABSTRACT

- [ ] Four sentences or fewer: the theme (identifying four metals by density, imprecise vs. precise tools), the measured densities, and the percent errors — nothing more

INTRODUCTION

- [ ] States $\rho = m/V$ and $V = \pi r^2 L$, with every variable described
- [ ] Background on how device precision sets uncertainty and significant digits
- [ ] Ends with the question the lab answers

EXPERIMENTAL

- [ ] Both tool sets described; three trials each of length, diameter and mass for all four rods, cycling through the measurements
- [ ] The thread holder for the spring scale, and how the Vernier scale was read
- [ ] A photo or a labeled sketch of the setup

RESULTS

- [ ] All four rods from both tool sets in numbered, captioned tables or figures, each introduced with “Table 1 shows…”
- [ ] Every measured value at the device's precision, with a matching uncertainty — caliper values carry more digits than ruler values
- [ ] Accepted densities stated, with their source
- [ ] One sample calculation each for an average, a volume, a density, and a percent error

DISCUSSION

- [ ] Opens with “We experimentally determined…”
- [ ] Identifies each rod's metal with its percent error and a stated confidence
- [ ] Applies the 10% rule correctly
- [ ] Compares the precision of the two tool sets using the actual numbers
- [ ] The phrase “human error” appears nowhere in the report

ADDITIONAL QUESTIONS

- [ ] All four questions (most precise tool set, best answer, time, worth it) worked into prose

CONCLUSION

- [ ] Restates densities and percent errors, names each rod's metal definitively, and suggests specific measurement improvements

REFERENCES

- [ ] Multiple sources in the numbered bracket format, including the density table you actually used

FORMAT

- [ ] Written on the Lab Template, in sentences and paragraphs throughout

Files for this lab

HANDOUT · PDF

This lab, printable

The whole page as a PDF. Print it and bring it to lab — the self-checks only work on this page, but everything else is there.

DATA SHEET · XLSX

Excel data sheet

Type your three trials into the white bordered cells, to the precision of the device. Averages, volume, density, the suggested metal and percent error are calculated for you. The accepted density goes in the yellow cell.

HANDOUT · PDF

Reading a Vernier caliper

How to read the fixed scale and the Vernier scale, with two practice readings. Read it before Part II.

GRADING

Lab 01 rubric

The rubric page, with Word and PDF copies. Exactly what your report is marked on, criterion by criterion. Read it before you write, not after — it doubles as an outline.

WRITING AID · DOCX

Example starter sentences

A model report with XX placeholders where your numbers go, and a prompt for every table and calculation the rubric asks for. Not a report to hand in — a way to start one.

REQUIRED · ON BRIGHTSPACE

Lab Template and Writing Lab Reports module

Not linked here — find them in the **Writing Lab Reports** module on Brightspace. Read both before you write.

References to get you started

Your references must include the source of your density table. Your textbook, this handout, and the Module 1 notes on significant figures all count. Use this format:

[1] [Authors' names], *Journal or Book title*, edition, Volume number, page number, (year).

Questions? Leave a comment on the Discussion Board first. Good luck! — Dr. Cho

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Adapted from the original Lab 1 by Prof. H. Trevor Johnson-Steigelman. Built 2026-10-30 from the Lab 01 manual (v3); uncertainty rules settled 2026-10-30; PHY 118 edition 2026-11-01.